

Student IDs Group P: 38705907 38868318 38709007 38802139 38770717

Shortages and the nature of the toilet paper industry

A toilet paper shortage occurred during COVID-19. The demand for toilet paper had exceeded the supply of it. People were starting to increase the time of staying at home, which means that the demand of using toilet paper would be also raised. However, this is not the exclusive reason of toilet paper shortage. The other reason is panic buying. Since the majority stayed at home, they felt anxious causing hysteria as shelves were empty.

“Stock cannot be supplemented timely” is a fact. There are some natures of toilet paper supply chain. Firstly, according to the article posted by Willy Shih on Forbes, toilet paper has low value density, because toilet paper is lightweight but is shipped in large volumes. The price of toilet paper is less than the cost of its transport. To save the cost, producers have domestic manufacturing to shorten the distance of shipment. Secondly, Willy Shih (2020) also said that people have a stable demand for toilet paper. The quantity of consumption of toilet paper in a region depends on the size of the population. The producers will produce based on the consumption pattern and will not produce more to increase the inventory, since the producers must increase the expenditure on storing the redundant products. Thirdly, the shipment of raw materials – wood fibre is long distance as the wood fibre comes from trees in the countries in the southern hemisphere. Due to the stable demand and long distance of shipment of raw material, producers also do not store extra raw material. Moreover, according to Lora Cecere (2021), the toilet paper industry commonly has multiple distribution channels, and toilet paper producers cannot immediately receive the signal to increase production.

There are 4 competitive priorities: Cost, Quality, Time, and Flexibility. Cost is one of the common competitions between brands. Brands can reduce the cost of a product to decrease the price of product, which means that they can compete with other brands. There are three ways to decrease the cost: costs of the raw materials, cost of shipment and cost of distribution channels. Taking Andrex toilet paper as an example: in Andrex official website, the detail page of Andrex classic clean toilet tissue (n.d.) is showing that the quantity of each roll has been increased 50% so that it can directly decrease shipment cost. Quality is the base of a product. It is the main consideration of some customers to choose which brand of products. Using Andrex classic clean toilet tissue again, it is introduced that it is used unique 3D wave technique which can have better clean with fewer sheets. It can be an effective competition point comparing with other goods. flexibility is a method to stabilize current market share or improve competitive advantages for new market share. It means that the flexibility of a product can flexibly satisfy demands of different groups of customers. For instance, Andrex classic clean toilet tissue (n.d.) is packaged using 30% recycled plastics. It can be attractive for the customers who think the environment is important, which is a good selling point for a specific group. Finally, Shipping time is “Time” in 4 competition priorities. Shops or retailers would not have any supply shortages if the shipment time is short. It is significantly important that timely supplements and delivery would not affect the profits of what retailers or shops can get. It means that retailers and shops would be willing to stock the products which have short shipping time. However, it is unachievable in toilet paper industry because of high shipment cost. Therefore, the industry should concentrate on 3 competitive priorities – cost, quality, and flexibility – to ensure the competitiveness in this huge market.

Why shortages occurred and were prolonged.

The nature of the product irrefutably prolonged and caused the shortage. Toilet paper itself is a good which is inexpensive yet lightweight and voluminous. As a result, it has a low value density (Shih, W 2020). An example of a good with a high value density is something like gold. Due to this property, it is not very tradeable across countries. Countries are self-sufficient in producing their own toilet paper due to this. It reduces the cost and demand to ship toilet paper. However, due to this, supermarkets were not able to keep up with demand. The relative cost of toilet paper is far lower than the cost of shipping and transporting it long distances. Therefore, during the shortage, it is a main reason that it was prolonged is the nature of the product as it is not worth being shipped across countries due to its low value density. A lack of shipping capability led to the shortage being prolonged as demand could not be satisfied this way.

The supply chain has 4 primary cycles within it. The customer order cycle, the replenishment cycle, manufacturing cycle and the procurement cycle. Due to the increase in demand, these cycles deviate from their original values. For example, the customer order cycle becomes larger as more people are ordering. The replenishment cycle occurs more often as they find it harder to meet demand. The manufacturing cycle becomes larger as they must produce more and so does the procurement cycle as they need more supplies. This disrupts the supply chain. It becomes more difficult to manage due to more uncertainty. Suppliers were therefore limited in their actions to overcome this prolonging the impact of the toilet paper shortage.

However, it could be argued that that is also due to the nature of the good. Demand is constant as toilet paper is a functional good. As a result, when demand shifted upwards so suddenly, manufacturers went out of stock too quickly and manufacturers didn't have ample time to respond. The reason demand has shifted so far was due to the properties of toilet paper. It is inexpensive and considered to be an essential good. As a result, it was susceptible to panic buying caused by the pandemic. It was further prolonged due to the variability of demand. With the variability of demand, it became harder for supply chains to predict demand and meet it, causing shortages to occur. Furthermore, stores only keep several weeks of stock in hand, which is partially responsible for the initial shortage.

The lack of capacity of the suppliers and manufacturers in the supply chain was also responsible for the shortage. As mentioned before, demand for toilet paper is usually stable, therefore, manufacturers and suppliers do not have large amounts of spare capacity as capacity utilization is quite high. Due to the lack cushion capacity, the excess demand was not met as quickly as it could have been thus prolonging the shortage. Furthermore, even if they were to try to increase their capacity, they would be met with difficulties. The factors of production are inelastic, meaning that even if the demand was to go up for toilet paper, it would take too long for them to increase their factors of production to meet it. For example, specialized machines for only home use and not commercial use are expensive and take a long time to receive the machinery (Cecere, L 2021). The characteristics and shortcomings of the supply chain itself did play a part in the shortage.

In conclusion, it was due to both the nature of toilet paper and the design of the supply chain. It was a culmination of these 2 factors which caused the shortage to occur and to be prolonged.

Demand shock, shortages and the bullwhip effect

During covid-19, the bullwhip effect occurred as toilet paper is not a seasonal consumer item, and toilet paper is something that has a steady demand in the market. During covid, the bullwhip effect was large. In normal times, manufacturers and suppliers can predict market demand more accurately. However, during covid-19, customers were buying and hoarding toilet paper, breaking the balance of supply and demand. There are many other important parts from the manufacturer to the final purchaser to form a complete

supply chain. The manufacturer estimates the supply of the production platform from the changes in the market, and each part will set up a safe storage capacity to minimize and avoid losses caused by product shortages. The huge demand shock and the shortage of goods have caused a lot of increase in the estimates of market demand by distributors and manufacturers, which has also led to manufacturers getting incorrect information. Manufacturers will speed up production to ensure that the crisis will reduce product shortages. At the same time, as the crisis passed, consumers hoarded a large amount of inventory, and it took them a while to get through, which meant that the storage cost of suppliers increased. This can lead to backlogs in stores and signal production cuts.

To solve this kind of problem, Professor Hau Lee produced a very good solution, he creates a kind of supply chain call triple a supply chain, Hau Lee says, a highly aligned and agile supply chain can ensure that manufacturers can cope with large demand shocks and product shortages in the short term.

Agility

1. It can promote the synchronous flow of real-time information among supply chain partners, so that suppliers and manufacturers do not need to estimate market demand due to special impacts caused by special circumstances.
2. Develop a reliable logistics system or partners. A reliable logistics system can effectively reduce transportation costs, so that enterprises can make better profits. If they have an independent logistics system, they can respond in time and make quick decisions under special circumstances Adjustment.
3. Formulate contingency plans and develop a crisis management team. In a special period like Covid-19, the management team can introduce policies like purchase restrictions to ensure that demand shocks will not affect the market estimates of manufacturers and suppliers. A bullwhip effect is less likely.

Adaptability

Adaptability is described as “the ability to adjust supply chain design to meet changes in market structure and to modify the supply network to reflect changes in strategy, technology and products.

1. Directly assess the needs of end customers and direct customers. In this way, when the enterprises in the middle part of the supply chain only make production or supply decisions based on the demand information from their adjacent subordinate enterprises, the untruth of the demand information will go upstream along the supply chain, resulting in the phenomenon of gradual amplification.
2. Determine the product's position in terms of technology and product life cycle.

Alignment

Alignment is described as the ability of a large company to "align the interests of all companies in its supply chain with its own".

1. Clearly stipulate the roles and responsibilities of suppliers and customers, such as the purchase

restriction policy. If in a special period, the behavior of customers hoarding goods and the loss of suppliers due to lack of goods have an impact on the manufacturer's assessment of the market, this can reduce the bullwhip effect.

2. Free exchange of information with suppliers and customers, free exchange of information means better frequency of communication with the market, more communication can allow manufacturers to understand the impact of demand caused by special periods and allow manufacturers to better respond to Evaluate the current market demand and make changes.

The portfolio approach

The portfolio approach to supply chain architecture entails reducing supply chain risk by diversifying the portfolio of goods, suppliers, and distribution channels (Olavson et al., 2010). This method acknowledges that a range of events, including natural catastrophes, economic downturns, geopolitical events, and supplier failures, can cause supply chain interruptions, by disrupting transportation network, or cause damage to facilities or inventory. It's possible that a single supply chain design won't be strong enough to withstand each danger. Consequently, by varying their supply chain designs, a portfolio strategy may assist businesses in risk management.

By adopting a more thorough and proactive approach to risk management, the portfolio approach to supply chain risk management expands on conventional risk management approaches. This method involves prioritizing and managing risks using a combination of risk mitigation, risk transfer, and risk acceptance tactics after analysing hazards in terms of their potential impact and likelihood. The supply chain can be described as a portfolio of assets, with each investment representing a particular risk- this is the approach taken with the portfolio approach. While some dangers may have a significant impact and a high possibility, others may not. The supply chain management can gain a more in-depth view of the supply chain's total risk profile by evaluating each risk in this manner.

The portfolio strategy, according to Olavson et al. (2010), entails creating a portfolio of supply chain architectures that differ in terms of their risk profile. This portfolio may contain many product categories, vendors, and routes of distribution, as well as various degrees of vertical integration. With riskier supply chain architectures offering higher potential rewards but also higher risk, the objective is to create a balance between risk and reward. reducing supply chain risk by diversifying the portfolio of goods, suppliers, and distribution channels, acknowledges that a range of events, including natural catastrophes, economic downturns, geopolitical events, and supplier failures, can cause supply chain interruptions. It's possible that a single supply chain design won't be strong enough to withstand each danger. Consequently, by varying their supply chain designs, a portfolio strategy may assist businesses in risk management. After identifying and classifying the various portfolios and assets, organizations should assess the risks in terms of their likelihood and impact before prioritizing them using a combination of risk acceptance, transfer, and mitigation measures.

The change in the supply chain of HP printers

In the 1990s HP printer engines were manufactured in Asia where costs were low and then assembled and packaged into specific stock keeping units (SKUs) in a country closer to where demand was concentrated (Olavson et al., 2010). Producing printer engines abroad is an example of outsourcing and more specifically offshoring. HP deliberately postponed the packaging of engines into different SKUs until demand for the

different SKUs was known (Olavson et al., 2010). This method is called postponement, which provided financial benefits for HP in the 1990s as it delivered customized products to customers at low costs. The unpredictable demand and high variety of HP printers shows it was an innovative product. The unpredictable demand caused HP to employ a responsive supply chain. The characteristics of responsive supply chains include high inventory and supplier choice based on speed, flexibility and quality. The supply chain of HP was optimized to meet consumer demand.

Over the years the design of the HP supply chain has changed. The number of supply chains has increased, and postponement is less prominent. The location of manufacturing has also changed, and now manufacturing and assembly occur in the same country. The supply chain now consists of a responsive supply chain (like that employed in the 1990s), a supply chain optimized for cost and a supply chain optimized for even lower costs (Olavson et al., 2010). Multiple supply chains are needed to accommodate the different costs and delivery times for different products. The move away from postponement was caused by competitive price pressures which necessitated much lower costs than postponement provided (Olavson et al., 2010). Price pressures were also the motive behind the migration to countries where production and assembly could occur at an even lower cost. This shows how offshoring is now even more prominent for HP.

Other market-driven factors caused changes in HP's supply chain. The printer market matured and so customer segments and market size were more defined and consequently fewer SKUs were needed (Olavson et al., 2010). These factors led to postponement no longer being necessary. Research and development in the market also led to improved product quality, consistency and longer life cycles. Consistent high-level production further reduces costs for repairs. All the price reductions contributed to price pressure which led to postponement becoming less economically viable for HP. Printers having reduced variety, more predictable demand, more consistent quality and a longer life cycle show how printers are becoming a more functional product. They still exhibit innovative characteristics however, e.g. high margins. The increased functional aspects of printers lead to supply chains being more efficient. Efficient supply chains exhibit low inventory and supplier choice is based on cost and quality. The supply chains are still a reflection of consumer demand as they were in the 1990s.

Changes in the supply chain were also caused by changing financial goals. In the 1990s when printers were still in their infant stage, the emphasis was placed on increasing market share and revenue, so HP employed a responsive supply chain. As the market matured the priority switched to profitability, which led to offshoring to countries with even lower costs, reduced SKUs and the abandonment of postponement due to its lower profit margin (Olavson et al., 2010). However, postponement still exists in the high-end printer market, where market share is still the priority and variety is needed. This shows why a responsive supply chain is necessary, and why the number of HP supply chains has increased since the 1990s to fit responsive and less responsive supply chains.

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